

DMMVIEW_G Communication Protocol

一. Specification of Communication Protocol

This Communication Protocol complies with SCPI.

Version: 0.0

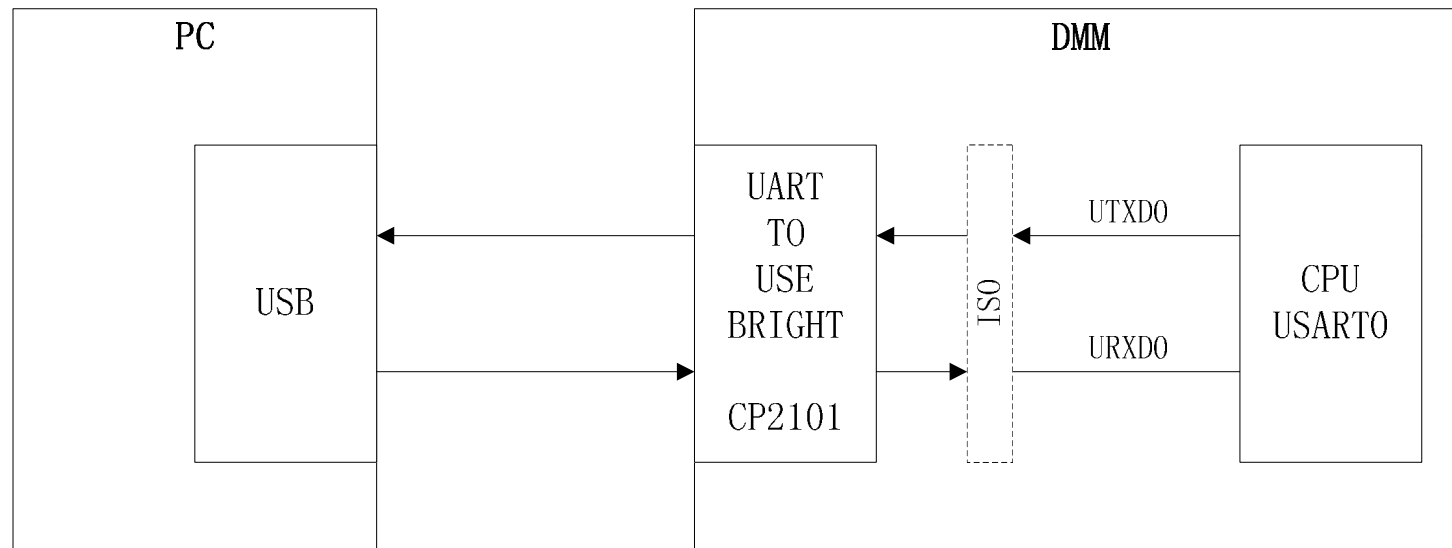
SCPI: Standard Commands for Programmable Instruments,

The reference of this Communication Protocol refers to:

agilent 34410A_11A_SCPI_Reference

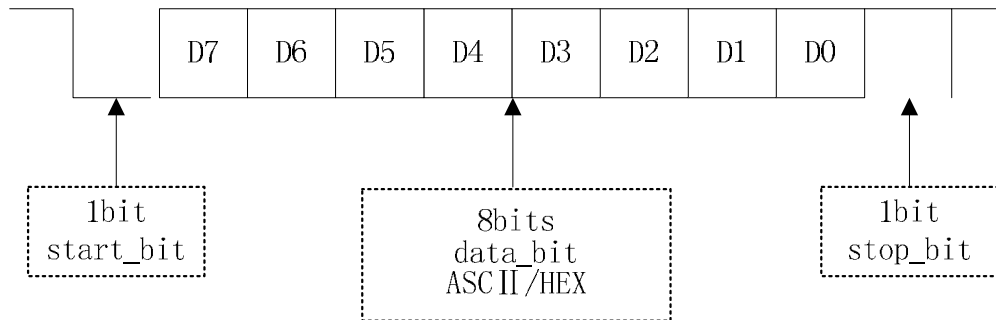
yokogawa 734 Digital Multimeter command Reference

二. Composition of Systematic Communication



- Full-duplex UART TO USB communication
- Communication baud rate: 9600 bps

- Point to point online communication
The upper PC is Master
DMM is Slaver
- Communication data frame format



- communication data package format
 - ✓ communication command format

HEAD	COMMAND	EM
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HEAD : 1 byte ASCII ‘#’

COMMAND: N bytes ASCII

EM : 2 bytes ASCII ‘CR LF’

✓ communication return data format:

HEAD	DATA	EM
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HEAD : 1 byte ASCII ‘#’

DATA : N bytes ASCII or HEX data

EM : 2 bytes ASCII ‘CR LF’

Communication Command Subset 1: basic command for online communication [BASE]

<RST> command	
Description	reset command
Command	#<RST><cr>
Returned Value	#<ACK><cr>
Specification	▪ Return back to ACK response. ▪ The Meter facilitates power-on operation, equivalent to pressing Power-on key.

<ONL>command	
Description	PC's online communication command
Command	#<ONL><cr>
Returned Value	#<ACK><cr>

Specification	▪ PC undertakes online communications with the Meter and enters into online communication work. ▪ Successful online, returning back to ACK response.
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Command Subset 2: the meter inputs measurement communication command [INPUT]

<RD?>command	
description	the meter inputs measured value and read commands(real-time)
command	#<RD?><cr>
returned value	▪ type 1: confirmed response #<RD> <u>x...x</u> <cr> See Appendix: 2, 5 x...x : measurement data, 26 bytes data ▪ type 2: non-confirmed response NAK #<NAK><cr>
specification	▪ The Meter enters into input measurement work, and read present measured data. Return back to RD response, and input measured data. ▪ If the meter is in recording reading state, then returns back to NAK response.

<SC?>command	
description	reading command to read manual recording data
command	#<SC?><cr>
returned value	▪ type 1: confirmed response #<SC> <u>x...x</u> <cr> x...x : number of manual recording data, 3 bytes ▪ type 2: non-confirmed response NAK #<NAK><cr>
specification	▪ The Meter enters into record reading work, reading present measured data. Return back to SC response, and record measured data. ▪ If the meter is in non-record reading state, then returns back to NAK response.

<SD?>command	
description	reading command to read manual recording data

command	#<SD?> <u>x...x</u> <cr> <u>x...x</u>: number of manual recording data, 3 bytes
returned value	▪ type 1: confirmed response #<SD> <u>x...x</u> <cr> See Appendix: 2, 5 x...x : measurement data, 26 bytes data ▪ type 2: non-confirmed response NAK #<NAK><cr>
specification	▪ The Meter enters into record reading work Return back to SD response, and ONE record data unit. ▪ If the meter is in non-record reading state, then returns back to NAK response.

<TC?> command	
description	reading command to read manual recording data
command	#<TC?><cr>

returned value	- type 1: confirmed response #<TC> <u>x...x</u> <cr> - x...x : number of real-time recording data, 3 bytes data - type 2: non-confirmed response NAK #<NAK><cr>
specification	- The Meter enters into record reading work to read real-time data work of the meter. Return back to TC response, and record data. - If the meter is in non-record reading state, then returns back to NAK response.

<TI?>Command	
description	reading command to read record interval time of the Meter
command	#<TI?><cr>
returned value	- type 1: confirmed response #<TI> <u>x...x</u> <cr> - x...x : number of interval time recording, 4 bytes data. - type 2: non-confirmed response NAK #<NAK><cr>

specification	▪ The Meter enters into record reading work to read interval time of the meter. Return back to TI response, and record data. ▪ If the meter is in non-record reading state, then returns back to NAK response.
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<TD?>Command	
description	reading command to read timing recording data
command	#<TD?> <u>x...x</u> <cr> <u>x...x</u> : timing recording numbers, 3 bytes data
returned value	▪ type 1: confirmed response #<TD> <u>x...x</u> <cr> See Appendix: 2, 5 x...x : timing measured data, 26 bytes data ▪ type 2: non-confirmed response NAK #<NAK><cr>
specification	▪ The Meter enters into record reading. Return back to TD response, and record ONE record data. ▪ If the meter is in non-record reading state, then returns back to NAK response.

Appendix 1: Instruction code

- specification of communication command code

instruction code	communication command
0x52 0x53 0x54	<RST>
0x4F 0x4E 0x4C	<ONL>
0x52 0x44 0x3F	<RD?>
0x53 0x43 0x3F	<SC?>
0x53 0x44 0x3F	<SD?>
0x54 0x43 0x3F	<TC?>
0x54 0x49 0x3F	<TI?>
0x54 0x44 0x3F	<TD?>

- specification of returning back to response code

instruction code	return command
0x06 0x00	<ACK>: confirmed response
0x15 0x00	<NAK>:non-confirmed response
0x52 0x44	<RD>: return back to measured data

0x53 0x43	<SC>: return back to manual recording data value
0x53 0x44	<SD>: return back to manual recording data
0x54 0x43	<TC>: return back to timing recording data value
0x54 0x49	<TI>: return back to recording interval time of timing recording
0x54 0x44	<TD>: return back to timing recording data

- **Specification of symbols in Protocol:**

Symbol	Meaning
?	inquire operation
<sp>	blank space symbol
x	on digit or decimals
x...x	variable digit or decimals
<cr>	Carriage Return
<lf>	line feed

Appendix 2: Specification of numbers of input measurement (function/ range) and structure of input measured data

basic measurement		number of measurement function	number of measurement range	structure of measured data
measured function	range			
AC voltage ACV	2V	0x30 0x30	0x30	±X.XXXX or FFFFFFFF
	20V		0x31	±XX.XXX or FFFFFFFF
	200V		0x32	±XXX.XX or FFFFFFFF
	1000V		0x33	±XXXX.X or FFFFFFFF
Decibel dBm_V	2V	0x30 0x31	0x30	±XX.XXX or FFFFFFFF
	20V		0x31	±XX.XXX or FFFFFFFF
	200V		0x32	±XX.XXX or FFFFFFFF
	1000V		0x33	±XX.XXX or FFFFFFFF
AC Ohm Voltage ACmV	200mV	0x30 0x32	0x30	±XXX.XX or FFFFFFFF
Decibel dBm_mV	200mV	0x30 0x33	0x30	±XX.XXX or FFFFFFFF
DC Voltage DCV	2V	0x30 0x34	0x30	±X.XXXX or FFFFFFFF
	20V		0x31	±XX.XXX or FFFFFFFF
	200V		0x32	±XXX.XX or FFFFFFFF
	1000V		0x33	±XXXX.X or FFFFFFFF
DC Ohm Voltage DCmV	200mV	0x30 0x35	0x30	±XXX.XX or FFFFFFFF
thermocouple TC	K	0x30 0x36	0x30	±XXXX.X or FFFFFFFF

continuity test CONT	600Ω	0x30 0x37	0x30	±XXXX.X or FFFFFFFF
Diode DIODE	2V	0x30 0x38	0x30	±X.XXXX or FFFFFFFF
Ohms OHM	200Ω	0x30 0x39	0x30	±XXX.XX or FFFFFFFF
	2kΩ		0x31	±X.XXXX or FFFFFFFF
	20kΩ		0x32	±XX.XXX or FFFFFFFF
	200kΩ		0x33	±XXX.XX or FFFFFFFF
	2MΩ		0x34	±X.XXXX or FFFFFFFF
	20MΩ		0x35	±XX.XXX or FFFFFFFF
	60MΩ		0x36	±XXX.XX or FFFFFFFF
thermal resistance RTD	PT100	0x31 0x30	0x30	±XXXXX. or FFFFFFFF
Capacity CAP	10nF	0x31 0x31	0x30	±XXX.XX or FFFFFFFF
	100nF		0x31	±XXXX.X or FFFFFFFF
	1000nF		0x32	±XXXXX. or FFFFFFFF
	10μF		0x33	±XXX.XXorFFFFFFFF
	100μF		0x34	±XXXX.XorFFFFFFFF
	1000μF		0x35	±XXXXX.orFFFFFFFF
	10mF		0x35	±XXX.XXorFFFFFFFF
	100mF		0x36	±XXXX.XorFFFFFFFF
Frequency FREQ Duty Cycle DUTY	10Hz	0x31 0x32	0x30	±X.XXXXorFFFFFFFF
	100Hz		0x31	±XX.XXXorFFFFFFFF
	1000Hz		0x32	±XXX.XXorFFFFFFFF
	10kHz		0x33	±X.XXXXorFFFFFFFF
	100kHz		0x34	±XX.XXXorFFFFFFFF
	1000kHz		0x35	±XXX.XXorFFFFFFFF
	10MHz		0x36	±X.XXXXorFFFFFFFF
	占空比		/	±XXX.XXorFFFFFFFF
DC Microamp Current DCuA	200uA	0x31 0x33	0x30	±XXX.XXorFFFFFFFF
	2000uA		0x31	±XXXX.XorFFFFFFFF
DC Milliampere Current	20mA	0x31 0x34	0x30	±XX.XXXorFFFFFFFF
	200mA		0x31	±XXX.XXorFFFFFFFF

DCmA				
Direct Current DCA	2A	0x31 0x35	0x30	±X.XXXXorFFFFFFFF
	10A		0x31	±XX.XXXorFFFFFFFF
AC Mictoamp Current ACuA	200uA	0x31 0x36	0x30	±XXX.XXorFFFFFFFF
	2000uA		0x31	±XXXX.XorFFFFFFFF
Milliampere Current ACmA	20mA	0x31 0x37	0x30	±XX.XXXorFFFFFFFF
	200mA		0x31	±XXX.XXorFFFFFFFF
AC Current ACA	2A	0x31 0x38	0x30	±X.XXXXorFFFFFFFF
	10A		0x31	±XX.XXXorFFFFFFFF
Peak Voltage PEAK_DCmV	200mV	0x31 0x39	0x30	±XXXX.XorFFFFFFFF
Peak Voltage PEAK_DCV	2V	0x32 0x30	0x30	±XX.XXXorFFFFFFFF
	20V		0x31	±XXX.XXorFFFFFFFF
	200V		0x32	±XXXX.XorFFFFFFFF
	1000V		0x33	±XXXXX.orFFFFFFFF
Peak Current PEAK_DCuA	200uA	0x32 0x31	0x30	±XXXX.XorFFFFFFFF
	2000uA		0x31	±XXXXX.orFFFFFFFF
Peak Current PEAK_DcmA	20mA	0x32 0x32	0x30	±XXX.XXorFFFFFFFF
	200mA		0x31	±XXXX.XorFFFFFFFF
Peak Current PEAK_DCA	2A	0x32 0x33	0x30	±XX.XXXorFFFFFFFF
	10A		0x31	±XXX.XXorFFFFFFFF
AC voltage plus frequency ACmV+HZ	10Hz	0x32 0x34	0x30	±X.XXXXorFFFFFFFF
	100Hz		0x31	±XX.XXXorFFFFFFFF
	1000Hz		0x32	±XXX.XXorFFFFFFFF
	10kHz		0x33	±X.XXXXorFFFFFFFF
	100kHz		0x34	±XX.XXXorFFFFFFFF
	1000kHz		0x35	±XXX.XXorFFFFFFFF
	10MHz		0x36	±X.XXXXorFFFFFFFF
	200mV		0x30	±XXX.XXorFFFFFFFF
AC voltage plus frequency ACV+HZ	2V	0x32 0x35	0x30	±X.XXXXorFFFFFFFF
	20V		0x31	±XX.XXXorFFFFFFFF
	200V		0x32	±XXX.XXorFFFFFFFF

	1000V		0x33	±XXXX.XorFFFFFFFF
AC current plus frequency ACuA+HZ	200uA	0x32 0x36	0x30	±XXX.XXorFFFFFFFF
	2000uA		0x31	±X.XXXXorFFFFFFFF
AC current plus frequency ACmA+HZ	20mA	0x32 0x37	0x30	±XX.XXXorFFFFFFFF
	200mA		0x31	±XXX.XXorFFFFFFFF
AC current plus frequency ACA+HZ	2A	0x32 0x38	0x30	±XXXX.XorFFFFFFFF
	10A		0x31	±XXX.XXorFFFFFFFF
AC current + DC voltage ACmV+DCmV	200mV	0x32 0x39	0x30	±XXX.XXorFFFFFFFF
AC current plus DC voltage ACV+DCV	2V	0x33 0x30	0x30	±X.XXXXorFFFFFFFF
	20V		0x31	±XX.XXXorFFFFFFFF
	200V		0x32	±XXX.XXorFFFFFFFF
	1000V		0x33	±XXXX.XorFFFFFFFF
AC plus DC current ACuA+DCuA	200uA	0x33 0x31	0x30	±XXX.XXorFFFFFFFF
	2000uA		0x31	±X.XXXXorFFFFFFFF
AC plus DC current ACmA+DCmA	20mA	0x33 0x32	0x32	±XX.XXXorFFFFFFFF
	200mA		0x33	±XXX.XXorFFFFFFFF
AC plus DC current ACA+DCA	2A	0x33 0x33	0x34	±X.XXXXorFFFFFFFF
	10A		0x35	±XX. XXXorFFFFFFFF
record reading MEM	/	0x33 0x34	/	/

Appendix 5: information structure of communication data unit

input measurement function (basic function)	Measurement Function (2byte)	Measurement data :1 Range : Rang1 (1byte)	Measurement data 1 Measured value :Data1 (7byte)	Measurement data 2 Range :Rang2 (1byte)	Measurement data 2 Measured value : Data2 (7byte)	Measurement data 3 Range Rang3 (1byte)	Measurement data 3 Measured value : Data3 (7byte)
ACV	0x30 0x30	0x30~0x33	ACV				
dBm_V	0x30 0x31	0x30~0x33	dBm	0x30~0x33	ACV		dBm_REF
ACmV	0x30 0x32	0x30	ACmV				
dBm_mV	0x30 0x33	0x30	dBm	0x30	ACmV		dBm_REF
DCV	0x30 0x34	0x30~0x33	DCV				
DCmV	0x30 0x35	0x30	DCmV				
TC_K	0x30 0x36	0x30	TEMP	0x30	DCmV	°C/°F	RJC
CONT	0x30 0x37	0x30	CONT				
DIODE	0x30 0x38	0x30	DIODE				
OHM	0x30 0x39	0x30~0x36	OHM				

RTD	0x31 0x30	0x30	TEMP	0x30	OHM	°C/°F	
CAP	0x31 0x31	0x30~0x37	CAP				
FREQ/DUTY	0x31 0x32	0x30~0x36	FREQ	/	DUTY		
DCuA	0x31 0x33	0x30~0x31	DCuA				
DCmA	0x31 0x34	0x30~0x31	DCmA				
DCA	0x31 0x35	0x30~0x31	DCA				
ACuA	0x31 0x36	0x30~0x31	ACuA				
ACmA	0x31 0x37	0x30~0x31	ACmA				
ACA	0x31 0x38	0x30~0x31	ACA				
PEAK_DCmV	0x31 0x39	0x30	PEAK_MAX	0x30	PEAK_MIN		
PEAK_DCV	0x32 0x30	0x30~0x33	PEAK_MAX	0x30~0x33	PEAK_MIN		
PEAK_DCuA	0x32 0x31	0x30~0x31	PEAK_MAX	0x30~0x31	PEAK_MIN		
PEAK_DCmA	0x32 0x32	0x30~0x31	PEAK_MAX	0x30~0x31	PEAK_MIN		
PEAK_DCA	0x32 0x33	0x30~0x31	PEAK_MAX	0x30~0x31	PEAK_MIN		
ACmV+HZ	0x32 0x34	0x30~0x36	FREQ	/	DUTY	0x30	ACV
ACV+HZ	0x32 0x35	0x30~0x36	FREQ	/	DUTY	0x30~0x33	ACV

ACuA+HZ	0x32 0x36	0x30~0x36	FREQ	/	DUTY	0x30~0x31	ACuA
ACmA+HZ	0x32 0x37	0x30~0x36	FREQ	/	DUTY	0x30~0x31	ACmA
ACA+HZ	0x32 0x38	0x30~0x36	FREQ	/	DUTY	0x30~0x31	ACA
ACV+DCmV	0x32 0x39	0x30	AC+DC	0x30	ACV	0x30	DCV
ACV+DCV	0x33 0x30	0x30~0x33	AC+DC	0x30~0x33	ACV	0x30~0x33	DCV
ACA+DCuA	0x33 0x31	0x30~0x31	AC+DC	0x30~0x31	ACI	0x30~0x31	DCuA
ACA+DCmA	0x33 0x32	0x30~0x31	AC+DC	0x30~0x31	ACI	0x30~0x31	DCmA
ACA+DCA	0x33 0x33	0x30~0x31	AC+DC	0x30~0x31	ACI	0x30~0x31	DCA